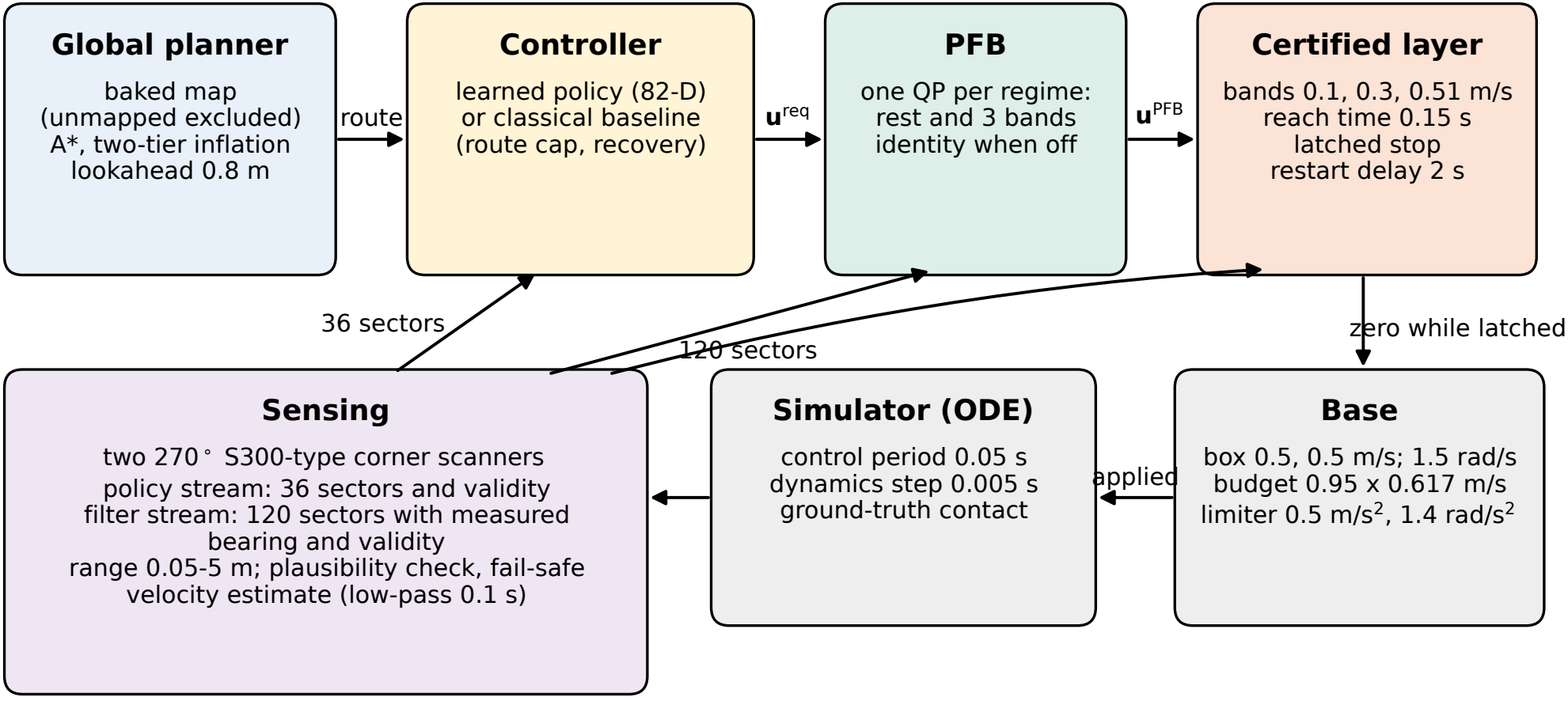
Global planner
baked map
(unmapped excluded)
A*, two-tier inflation
lookahead 0.8 m
route
Controller
learned policy (82-D)
or classical baseline
(route cap, recovery)
u^req
PFB
one QP per regime:
rest and 3 bands
identity when off
u^PFB
Certified layer
bands 0.1, 0.3, 0.51 m/s
reach time 0.15 s
latched stop
restart delay 2 s
zero while latched
Base
box 0.5, 0.5 m/s; 1.5 rad/s
budget 0.95 x 0.617 m/s
limiter 0.5 m/s², 1.4 rad/s²
applied
Simulator (ODE)
control period 0.05 s
dynamics step 0.005 s
ground-truth contact
Sensing
two 270° S300-type corner scanners
policy stream: 36 sectors and validity
filter stream: 120 sectors with measured
bearing and validity
range 0.05-5 m; plausibility check, fail-safe
velocity estimate (low-pass 0.1 s)
36 sectors
120 sectors